

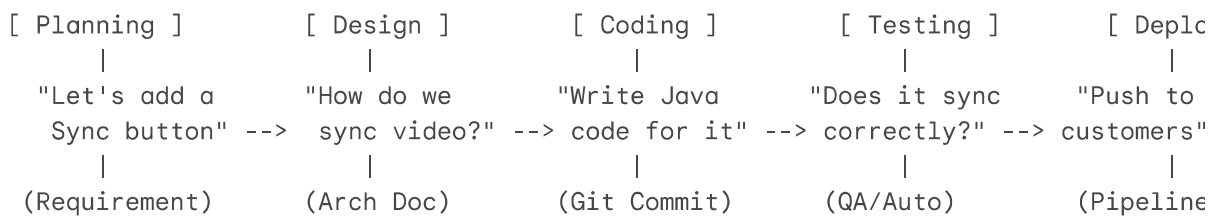
Amazon Internship: Software Testing & SDLC Guide

Target Audience: 2nd Year SDE / QA / SDET Candidates **Focus:** SDLC, STLC, Testing Levels, and Amazon Purchase Workflows.

1. SDLC (Software Development Life Cycle)

SDLC is the framework defining tasks performed at each step in the software development process. At Amazon, this process is highly iterative and customer-centric.

SDLC Purchase Workflow Diagram



Detailed Stage-by-Stage Breakdown:

1. Requirement Analysis (The "What"):

- **Process:** Product Managers (PMs) gather customer feedback. At Amazon, this often starts with a **PR/FAQ** (a mock Press Release) to visualize the finished product.
- **Example:** For "Watch Party," the requirement might be: "Users must see the exact same frame of the movie within 100ms of each other."

2. Design (The "How"):

- **Process:** SDEs create a **Design Document**. This covers high-level architecture (which AWS services to use) and low-level details (class structures).
- **Example:** Deciding to use *Amazon ElastiCache* to store the 'current timestamp' of the video for the host.

3. Implementation (Coding):

- **Process:** Writing the actual source code. Amazon follows **Clean Code** principles and uses internal version control (similar to Git).
- **Example:** Writing a Java service that triggers a `pause()` event to all connected clients when the host clicks the button.

4. Testing (Verification):

- **Process:** This isn't just "finding bugs"; it's ensuring the software meets the PRD requirements.
- **Example:** Simulating high-latency network conditions (slow internet) to see if the sync logic holds up.

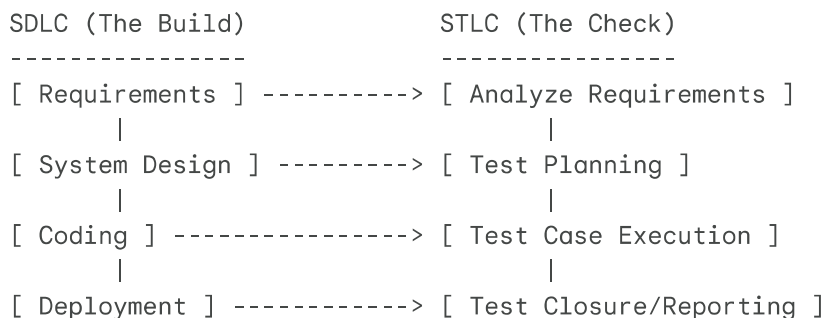
5. Deployment (Release):

- **Process:** Moving code to production. Amazon uses **CI/CD (Continuous Integration/Continuous Deployment)**, meaning code is automatically tested and shipped multiple times a day.

2. **STLC (Software Testing Life Cycle)**

While SDLC builds the car, STLC is the rigorous safety testing of every nut and bolt.

Parallel Life Cycles (V-Model Concept)



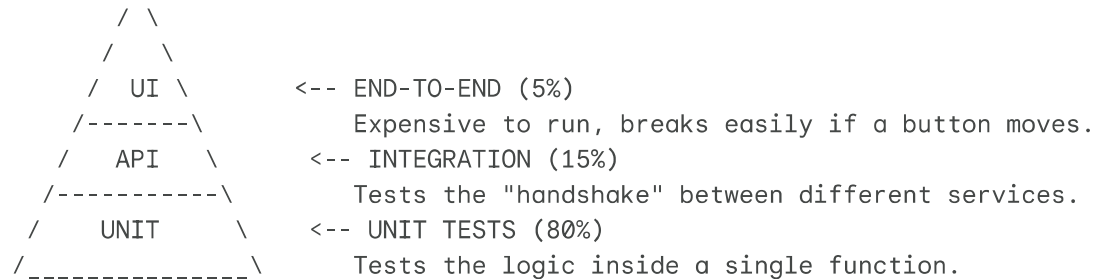
STLC Phases Deep-Dive:

- **Requirement Analysis:** Testers review the requirements to see if they are **testable**.
 - *Example:* If a requirement says "The map should load fast," the tester asks, "What is 'fast'? Is it 2 seconds or 500ms?"
- **Test Planning:** Defining the **Test Strategy**. This includes resource allocation (who is testing what) and tool selection (using *Selenium* or *Appium*).
- **Test Case Development:** Writing detailed steps. A good test case includes **Pre-conditions**, **Steps**, and **Expected Result**.
 - *Amazon Fresh Example:*
 - *Pre-condition:* User is a Prime Member.
 - *Action:* Add \$35 of groceries.
 - *Expected:* Shipping fee is waived at checkout.

- **Test Environment Setup:** Setting up a "staging" or "gamma" environment that mimics the real Amazon store so tests don't mess up real customer data.
- **Test Execution & Bug Reporting:** Running scripts. If **Actual Result** $\neq$ **Expected Result**, a bug is logged in a tracker (like Amazon's internal *SIM* tool).

3. 🚩 The Testing Pyramid (Amazon Strategy)

Amazon follows the Pyramid to ensure that the most numerous tests are also the fastest and cheapest.



- **Why 80% Unit Tests?** If a `CalculateTotal()` function has a math error, a Unit test finds it in 0.01 seconds. If you wait for a UI test, you have to wait for the browser to load, the user to log in, and the cart to open—wasting time and money.

4. 🛠️ Testing Techniques: Logic Visuals

A. Boundary Value Analysis (BVA)

Bugs usually hide at the "edges" of input ranges. This technique focuses on testing the minimum, maximum, and just outside those boundaries.

Scenario: Amazon "Free Shipping" starts at \$35.00.

Invalid (\$)		Valid (Free Shipping)			Valid	
----- -----		----- -----			-----	
[34.98]	[34.99]	[35.00]	[35.01]	[35.02]		[100.00]
(FEE)	(FEE)	(FREE)	(FREE)	(FREE)		(FREE)
		^				
		THE BOUNDARY				

- **The Logic:** Developers often use `>=` (greater than or equal to) or `>` (greater than). A common "off-by-one" error might make the free shipping start at \$35.01 instead of \$35.00.

Testing these exact three points (\$34.99, \$35.00, \$35.01) catches this.

B. Equivalence Partitioning (EP)

Instead of testing 1,000 different values, you group inputs that should behave the same way.

Scenario: Amazon Gift Card values (Accepts \$10 to \$2000).

Partition 1 [Too Low]	Partition 2 [Valid]	Partition 3 [Too High]
----- (-∞ to \$9.99)	----- (\$10 to \$2000)	----- (\$2001 to +∞)
Test: [\$5]	Test: [\$500]	Test: [\$5000]
Result: Reject	Result: Accept	Result: Reject

- **The Logic:** If the system correctly processes a \$500 gift card, it is highly likely it will process a \$600 or \$700 card. Testing one value per "bucket" saves time while ensuring coverage.

5. 🚀 Amazon "Quality Culture" Concepts

Canary Deployment: The "Blast Radius"

The "Canary" is a small group of users who receive a new update first.

- **The Workflow:**
 1. 95% of traffic stays on the old, safe version.
 2. 5% of traffic is routed to the new version.
 3. Automated monitors check for "404 Errors" or "Latency Spikes."
 4. If the 5% experience issues, the update is **instantly rolled back**.
- **Benefit:** Only a few people see the bug, preventing a global Amazon outage.

Regression Testing: "Safety First"

- **Definition:** Re-testing unchanged parts of the system.
- **Example:** Amazon updates the "Search Bar" UI. Regression testing ensures that this update didn't accidentally break the "Add to Cart" button, even though they seem unrelated.

📝 Practice MCQs (For Amazon Prep)

Q1: In the SDLC, which document describes "How" a system will work?

- A) PRD (Product Requirement Document)
- B) Design Document
- C) User Manual
- D) Bug Report

Answer: B. Explanation: While a PRD says *what* the customer wants, the Design Document is the engineering blueprint explaining architecture and logic.

Q2: What is "Regression Testing"?

- A) Testing a feature to see how much load it can handle.
- B) Re-testing unchanged parts of software after a fix to ensure no new bugs were introduced.
- C) Testing software in a different language.
- D) Testing without a plan.

Answer: B. Explanation: Think of it as "don't break the past." Whenever you add something new, you run old tests to make sure old features still work.

Q3: You are testing a password field that must be 8-16 characters long. Using BVA, which lengths should you test?

- A) 8, 12, 16
- B) 7, 8, 16, 17
- C) 1, 8, 16, 20
- D) 0, 5, 10, 15

Answer: B. Explanation: BVA checks the boundary (8, 16) and just outside (7, 17) to catch "off-by-one" errors.

Q4: Which layer of the Testing Pyramid catches bugs most cheaply?

- A) UI Tests
- B) Manual Tests
- C) Unit Tests
- D) Beta Tests

Answer: C. Explanation: Unit tests run in milliseconds during development. Catching a bug here costs pennies compared to catching it in production.

Q5: What is "Sanity Testing"?

- A) Testing to see if the tester is sane.
- B) A quick, narrow check to ensure a specific bug fix works as expected.
- C) Testing the entire application from scratch.
- D) Testing the app on a mobile device.

Answer: B. Explanation: Sanity testing is a subset of regression testing—a quick check to see if specific functionality is working after a change.

Q6: If a bug has "High Severity" but "Low Priority," what does it mean?

- A) It crashes the app, but only in a very rare, old browser.
- B) It's a typo on the home page.
- C) The app is slightly slow for everyone.
- D) The login button is missing.

Answer: A. Explanation: Severity is technical impact (Crash = High). Priority is business urgency. If no one uses that browser, we don't fix it immediately.

Q7: What is "Black Box Testing"?

- A) Testing internal code structure.
- B) Testing software without knowing its internal code or logic.
- C) Testing hardware components.
- D) Testing data center power.

Answer: B. Explanation: You only care about Inputs/Outputs, treating software like a "Black Box" you can't see into.

Q8: During which STLC phase is the "Traceability Matrix" (RTM) created?

- A) Test Planning
- B) Requirement Analysis
- C) Environment Setup

D) Test Closure

Answer: B. Explanation: The RTM maps requirements to test cases. You start this early to ensure 100% test coverage.

Q9: What is "Smoke Testing"?

A) Testing if the computer catches fire.

B) A high-level check to see if critical functionalities work (e.g., Can the app start?).

C) Detailed testing of a specific module.

D) Testing UI colors.

Answer: B. Explanation: If you turn a machine on and it smokes, you don't test features—you shut it down.

Q10: What is the benefit of a "Canary Deployment"?

A) It is faster than normal deployment.

B) It requires no testing.

C) It limits the "Blast Radius" of a bug to a small group of users.

D) It allows for better UI design.

Answer: C. Explanation: It's a safety mechanism to ensure new code doesn't affect millions of shoppers at once.

Q11: Which methodology is Amazon most likely to use for rapid feature releases?

A) Waterfall

B) Agile/Scrum

C) Big Bang

D) Spiral

Answer: B. Explanation: Agile allows for iterative development and frequent releases, fitting Amazon's "High Velocity" culture.

Q12: What is "Equivalence Partitioning" used for?

A) To test every single possible input value.

B) To reduce the number of test cases while maintaining good coverage.

- C) To test database speed.
- D) To find design document bugs.

Answer: B. Explanation: It groups similar inputs so you only test one "representative" from each group.

Q13: In STLC, what does "Test Execution" involve?

- A) Writing test cases.
- B) Running scripts and comparing Actual vs. Expected results.
- C) Designing the database.
- D) Planning the project budget.

Answer: B. Explanation: Execution is the phase where you run tests and log bugs if Actual ≠ Expected.

Q14: What is the "Actual Result" in a bug report?

- A) What software *should* do.
- B) What software *actually* did during the test.
- C) Steps to reproduce the bug.
- D) Name of the tester.

Answer: B. Explanation: If the button should be Blue (Expected) but it is Red (Actual), that mismatch is the bug.

Q15: Why is "Unit Testing" usually done by Developers?

- A) Because QAs don't know how to code.
- B) Because it requires knowledge of internal code (White Box).
- C) Because it is the most boring part.
- D) Because Unit tests aren't important.

Answer: B. Explanation: Since Unit tests check logic inside the code, the person who wrote the